

Pragya Sharma

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SUMMARY

Data Science graduate with strong academic and project-based experience in data analysis, machine learning, and applied AI, including NLP and generative models. Skilled in delivering data-driven insights and AI-powered solutions using Python, SQL, AWS SageMaker, Bedrock, and Airflow, with hands-on experience in statistical analysis, A/B testing, causal inference, regression, time-series, and text analytics. Experienced in building end-to-end analytical workflows and collaborating across teams to deliver scalable, actionable solutions.

EDUCATION

University of Massachusetts Dartmouth — MS in Data Science, North Dartmouth MA Sept 2023 - Dec 2025
Coursework - Adv. Machine Learning, Statistical Modeling, Adv. Data Mining GPA: 3.77
SVKM's NMIMS — B.Tech in Computer Science and Business System, India July 2019 - Aug 2023

TECHNICAL SKILLS

Language and Tools: Python, SQL, NoSQL, R, MATLAB, Git
ML/Stat Techniques: Regression, Causal Inference, Clustering, Time-Series, A/B Testing, Supervised/Unsupervised Learning
Frameworks Platforms: AWS (SageMaker, S3, Lambda), GCP, Snowflake, Airflow
Libraries: Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn
Visualization Tools: Power BI, Tableau, Looker
Certifications: AWS Cloud Practitioner Essentials, Generative AI: Introduction to Large Language Models (LLM)

EXPERIENCE

Machine Learning and Data Science Intern — Feynn Labs Services June 2024 – Aug 2024

- Increased **roommate compatibility success rates by 35%** and **reduced search time by 50%** by designing and deploying **ML-driven matching algorithms** for urban housing markets using **predictive analysis**.
- Spearheaded data pipelines and performed economic segmentation analysis using **GLMs and XGBoost** to deliver strategic recommendations that influenced **20%** of the company's market expansion decisions.
- Engineered **business and financial models** for an **AI-powered smart search platform**, improving **online purchase recommendation accuracy by 30%** adhering to **data compliance** standards.

PROJECTS

AI-Powered Research Paper Tool Capstone Project Jan 2025 - Dec 2025

- Built an **AI-driven research paper tool** with **enhancement features** using LLM APIs and NLP.
- Implemented **structured, section-wise extraction and summarization** of PDFs for faster research analysis.
- Deployed a **web-based system with backend APIs**, using cloud services and GitHub.

AutoGPT Implementation with Email Automation April 2025 - May 2025

- Fine-tuned and deployed **AutoGPT (v0.3.1)** using **Git**, and **OpenAI API** for autonomous multi-step task execution.
- Integrated **SMTP/IMAP Gmail plugin** to build **AI-powered email automation** system for live communication.
- Improved plugin management and dependency handling to enable seamless, low-intervention workflows.

Telecommunication Customer Churn Analysis March 2025 - April 2025

- Identified churn prediction accuracy using **ad-hoc analysis, XGBoost, logistic regression** and **random forest**.
- Achieved **80%** model performance via **GLM** and **multi-fact** data models for lifetime value prediction.
- Delivered retention insights by analyzing churn behavior and economic indicators, enhancing business understanding.

Fraud Detection in Financial Transactions Jan 2025 - Feb 2025

- Built an **ML security system** with **Scikit-Learn**, reaching **98%** fraud detection accuracy across 1,000+ clients.
- Used **SMOTE** to balance 9:1 class ratio, improving recall by **25%**, and fine-tuned **decision tree** to reach **98%** precision.
- Analyzed historical transaction data to refine fraud detection, cutting false positives by **30%** and boosting efficiency.

Article Text Extraction and Analysis Nov 2024 - Jan 2025

- Scraped and processed 100+ news articles using Python and **BeautifulSoup** for automated content extraction.
- Applied **NLP** techniques to compute sentiment scores and readability metrics for unstructured text.
- Exported structured insights to **Excel**, enabling efficient analysis and streamlined reporting.

Handwriting Recognition Nov 2024 - Dec 2024

- Built handwriting recognition model using custom **CNN** in **TensorFlow**, achieving **92%** accuracy on MNIST dataset.
- Designed a **CNN architecture** with dropout and essential layers to improve model generalization.
- Applied **deep learning** and **computer vision** to real-world classification task using efficient model architecture and training.